A Physics-Constrained Deep Learning Framework for Data-Efficient Density Current Prediction in Fluid Systems

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***Abstract*—Density currents are buoyancy-driven flows that occur when fluids of different densities meet, creating spontaneous motion without external forcing. These flows exhibit complex structures—sharp interfaces, unstable waves, and multi-scale vortices—fundamental to hydraulic engineering, environmental management, and climate modeling. Traditional theoretical models oversimplify three-dimensional turbulent structures, while high-resolution numerical simulations demand excessive computational resources for large-scale applications. This study integrates four stages: lock-exchange tank experiments with Planar Laser-Induced Fluorescence (PLIF) and Particle Image Velocimetry (PIV) measurements provide coupled density-velocity data; theoretical derivation establishes governing equations; high-resolution computational fluid dynamics (CFD) simulations generate detailed flow data; and PINN-LOCK reconstructs velocity, pressure, and salinity fields. PINN-LOCK achieved a Root Mean Squared Error (RMSE) of 0.089 for SUNTANS field reconstruction and a RMSE of 0.076 for the noisier experimental field. The model accurately reproduces interfacial instabilities and vortex dynamics at a fraction of CFD computational cost, achieving a higher accuracy under sparse data compared to traditional interpolation methods.**

***Index Terms*—density current, lock-exchange, simulation, PINNs, composite loss function**

I. Introduction

Density currents, also known as gravity currents, are buoyancy-driven flows that arise wherever fluids of different densities meet. They range from cold air sinking in a room, to the collision of atmospheric air masses that produces strong winds and heavy rainfall, to the thermohaline circulation that regulates Earth's climate [1], to turbidity currents and saline intrusions in stratified water bodies [2], [3]. They are therefore a central subject in oceanography, meteorology, and environmental engineering. Yet these flows are strongly unsteady, nonlinear, and three-dimensional, which makes direct measurement in natural environments nearly impossible and laboratory observations both expensive and spatially limited. This persistent scarcity of observational data raises a concrete computational question: how much data is actually required for a learning-based model to faithfully reconstruct a density current?

The lock-exchange experiment is the canonical laboratory model for studying such flows [4]. A closed tank is divided by a vertical partition into two compartments holding fluids of different densities; upon removal of the partition, two counter-propagating gravity currents develop along the top and bottom boundaries, accompanied by interfacial mixing, shear instabilities, and vortex formation [5]. Because it is well-characterised both experimentally and numerically, it provides an ideal controlled testbed for a data-efficiency study.

The governing equations for such flows are the incompressible Navier–Stokes equations, derived from Newton's second law [6]:

$$\frac{\partial u}{\partial t} + u \cdot \nabla u = -\frac{1}{\rho}\nabla p + \nu \nabla^2 u + f, \tag{1}$$

where u is the velocity field, ρ the density, p the pressure, ν the kinematic viscosity, and f an external body force. Their nonlinearity precludes analytical solutions, so researchers have relied on numerical methods [7]. Achieving high fidelity typically demands substantial computational resources [8], while experimental measurements remain sparse and noisy. This motivates the search for partial differential equation (PDE) solvers that combine efficiency with the ability to extract high-resolution flow features from imperfect data.

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [9], offer one such approach. PINNs embed PDE residuals as soft constraints in the loss function of a neural network, so that the governing equations act as a form of physical supervision in addition to any available observational data. Subsequent work [10], [11] has highlighted data efficiency as one of their defining advantages: because the PDE residual is evaluated by automatic differentiation at arbitrary collocation points, it can in principle compensate for very sparse measurements. Nevertheless, the quantitative relationship between training-data density and predictive accuracy is under-studied, particularly for unsteady flows with sharp internal structures.

To answer it, the study combines lock-exchange tank experiments, theoretical analysis, high-resolution numerical simulation, and deep learning. A PINN (hereafter *PINN-LOCK*) is trained on controlled sparse subsets of both CFD ground-truth data and PIV/PLIF experimental measurements, and its accuracy is evaluated across flow variables as the fraction of available training data is varied.

II. METHODS

A. PIV/PLIF Measurements

Particle Image Velocimetry/Planar Laser-Induced Fluorescence (PIV/PLIF) is a transient, multi-point, and non-intrusive technique for measuring fluid velocities and concentration [12]. In this method, tracer particles with excellent tracking and reflective properties are seeded into the flow. A laser sheet illuminates a cross-sectional plane of the flow, and an imaging system captures two or more exposures of the particle field. By analyzing the captured images using cross-correlation techniques, the average displacement of particles within each small sub-region is determined, enabling the reconstruction of the two-dimensional velocity distribution across the flow cross-section.

The overall experimental setup and its connections are shown in Fig. 1, comprising two cameras, a light source, a host computer, and the experimental rig with tank. The primary goal of PIV measurements is to capture the motion of the fluid by tracking the displacement of particles between two consecutive images. To measure density, fluorescent substances were put into the side with higher density. PLIF records laser-induced fluorescence within an instantaneous planar region of the flow. PLIF camera captured only the tracer data in the experiment while ignoring the seeding particles used for the PIV measurements.

The laboratory experiments were conducted in a two-dimensional tank with 100 cm in length, 10 cm in width, and 25 cm in height, with a total water depth of 20 cm for this experiment. In this experiment, the initial density difference is 0.005 g/cm^3. The measured maximum velocity is about 0.1 m/s occurring at the front. The density current exhibits a clear interface. As the relative velocity between the two fluid layers increases, the interface becomes unstable and begins to oscillate. This process forms a transition layer in the vertical direction, where both density and velocity gradually vary, resulting in distinct density and velocity gradients. Overall, the flow field obtained from the lock-exchange experiment exhibits highly unsteady and nonlinear behavior.

B. Theoretical Analysis

To better understand the dynamics of density currents, a theoretical solution is derived following the approach of Benjamin (1968) [13], employing the control volume method. Assume a flume is divided into left and right sides by a partition. The left half contains denser fluid (ρ_2), and the right half contains lighter fluid (ρ_1). We define a control volume moving with the front of the dense current.

Assuming steady-state energy flux without dissipation, we obtain ${U_f}^2 = 2{u_2}^2$.

$$ {U_f}^2 = g'h\left(1 - \frac{h}{H}\right) = g_0\frac{\Delta\rho}{\rho_0}h\left(1 - \frac{h}{H}\right). \tag{2} $$

The theoretical derivation based on the control volume method shows that the front velocity of the gravity current is proportional to the square root of the reduced gravity acceleration. It should be noted that this theoretical result is derived under idealized assumptions—*inviscid flow, no entrainment, no bottom friction, and steady-state conditions*—and thus does not account for turbulent mixing or unsteady effects. Consequently, deviations from experimental observations are expected.

C. Numerical Simulation

Numerical simulation provides a powerful means of investigating such complex fluid dynamics. For Lock-exchange flows, computational fluid dynamics (CFD) methods [14] are typically used to solve the full *Navier–Stokes equations*. Unlike analytical models, which rely on strong simplifications, numerical solvers based on discrete conservation laws can resolve complete spatiotemporal fields of velocity (u, w), pressure (p), and density (ρ).

We employed the *Stanford Unstructured Nonhydrostatic Terrain-following Adaptive Navier–Stokes Simulator* (SUNTANS) [15] to simulate Lock-exchange gravity currents. In SUNTANS, the horizontal mesh is generated using Delaunay triangulation, and the vertical grid employs a structured z-layer scheme. The computational domain extends 100 m in the x direction (L) with 200 cells, and 20 m in depth (z) with 50 vertical layers. For the boundary conditions, all boundaries except for the free surface are closed, meaning they are solid walls with zero velocity $(u, w = 0)$. The time step is set to $\Delta t = 0.2$ s, and the simulation is run for a total of 24,000 steps.

D. PINN-LOCK

To model this gravity current, we construct a physics-informed neural network, termed *PINN-LOCK*, designed to learn the spatiotemporal distributions of velocity components (u, w), pressure p, and salinity s. The model employs an explicit composite loss function that jointly enforces the Navier–Stokes momentum equations, the continuity equation, the scalar transport equation, as well as data, initial, and boundary conditions.

The central goal of the *PINN-LOCK* model is to approximate the following mapping with a deep neural network:

$$ (u, w, s, p) = N_\theta(x, z, t), \tag{3} $$

where θ represents the trainable parameters. The network must not only fit observational data but also satisfy the physical constraints imposed by the momentum, continuity, and salinity transport equations. To this end, the model employs a composite loss function that includes four residual terms (f, g, h, c). By minimizing this loss, the model jointly optimizes for both data fidelity and physical consistency.

The neural network adopts a fully connected feed-forward architecture with configurable depth and width. The input is a three-dimensional vector composed of the horizontal position x, vertical position z, and time coordinate t. The training data are derived from the SUNTANS numerical simulation results, comprising 151 temporal frames. Each time step contains 10,000 spatial sampling points, resulting in approximately 1.51 million data points in total. All field variables were

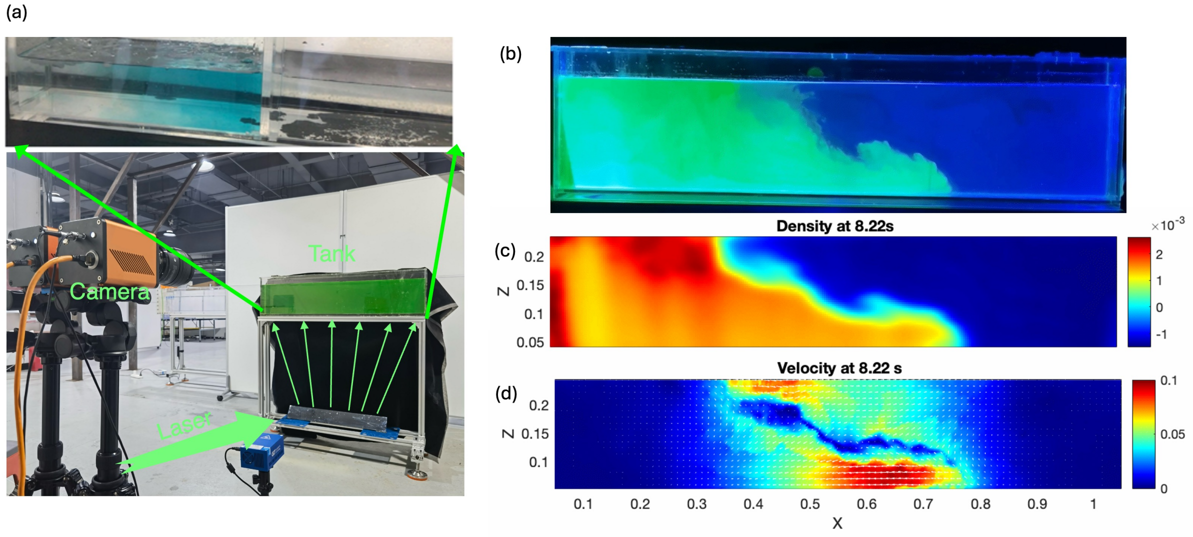

Fig. 1. Experimental setup of the PIV/PLIF system. (a) Lock-exchange tank with the PIV/PLIF system; (b) water with tracer particles; (c) PLIF measured density (density is shown as the density difference, g/cm^3); (d) PIV measured velocity field.

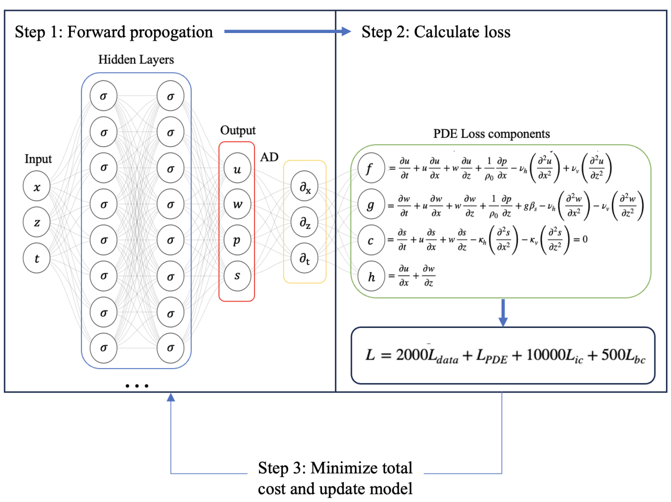

Fig. 2. Overall structure of the *PINN-LOCK* framework, illustrating the computation and iterative update of the total loss.

normalized using Z-score standardization to ensure balanced gradient propagation during optimization.

The network architecture begins with an input layer that linearly maps the three-dimensional coordinates to a configurable hidden dimension (128 neurons). The initial layer employs the Gaussian Error Linear Unit (GELU) activation function to ensure smooth differentiability of gradients, which is essential for accurate residual computation. Six subsequent hidden layers use the hyperbolic tangent (tanh) activation function. The final output layer consists of four neurons corresponding to the horizontal velocity u, vertical velocity w, pressure p, and salinity s (Fig. 2).

Physical constraints in *PINN-LOCK* are enforced via the residuals of the governing partial differential equations (PDEs), defined as:

$$\begin{aligned} L_{\text{PDE}} = \; & \text{MSE}(f, 0) + \text{MSE}(g, 0) \\ & + \text{MSE}(h, 0) + \text{MSE}(c, 0), \end{aligned} \tag{4}$$

These residuals are computed through automatic differentiation during forward propagation. The Mean Squared Error (MSE) formulation drives the residuals toward zero, embedding the Navier–Stokes equations directly into the optimization process.

The data loss L_{data} is defined as the MSE between predictions and the corresponding ground truth. The initial condition is strictly enforced at the earliest time step ($t = 0$), corresponding to the static, stratified state prior to flow onset. The boundary conditions correspond to the impermeable walls of the rectangular domain:

$$\begin{aligned} L_{\text{data}} = \; & \text{MSE}(u, u_{\text{pred}}) + \text{MSE}(w, w_{\text{pred}}) \\ & + \text{MSE}(p, p_{\text{pred}}) + \text{MSE}(s, s_{\text{pred}}), \\ L_{\text{ic}} = \; & \text{MSE}(u, u_0) + \text{MSE}(w, w_0) \\ & + \text{MSE}(p, p_0) + \text{MSE}(s, s_0), \\ L_{\text{bc}} = \; & \text{MSE}(u|_{x \leq 0}, 0) + \text{MSE}(u|_{x \geq 100}, 0) \\ & + \text{MSE}(w|_{z \leq 0}, 0) + \text{MSE}(w|_{z \geq 20}, 0). \end{aligned} \tag{5}$$

The total loss integrates all components:

$$L = 2000L_{\text{data}} + L_{\text{PDE}} + 10000L_{\text{ic}} + 500L_{\text{bc}}. \tag{6}$$

Training is conducted using the L-BFGS optimizer, enhanced with a strong Wolfe line search to promote efficient convergence. The network is trained for a total of 500 epochs.

In this study, two types of datasets are used to train the *PINN-LOCK*: synthetic data generated by numerical models SUNTANS and experimental observations obtained from PIV and PLIF measurements. Model data are complete and noise-free, allowing the network to primarily validate the governing

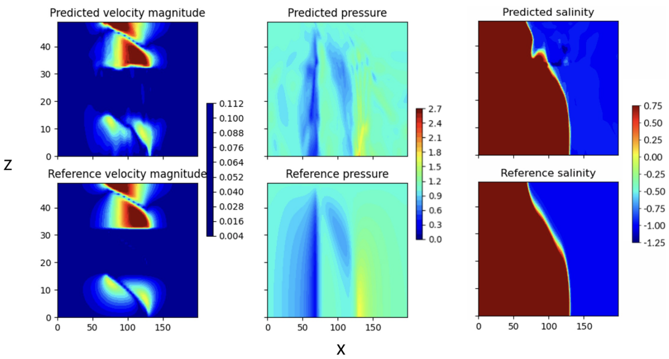

Fig. 3. Spatial distributions at $t = 300$.

PDEs and reproduce physically consistent solutions. Observational data, by contrast, are sparse and noisy, often missing key variables. Training with such data requires stronger PDE constraints, enabling denoising, inferring unmeasured quantities, and reconstructing the full flow field. Thus, model data serve as controlled benchmarks, while observational data demonstrate PINN's practical value.

III. Results

A. SUNTANS Data

The predictive performance of the *PINN-LOCK* model is evaluated against reference solutions obtained from SUNTANS numerical simulations, focusing on velocity magnitude, salinity, and pressure fields. All quantities presented in the following figures are normalized for consistency.

At the mid-stage ($t = 300$), the reference field develops more complex features, including vortex roll-up and entrainment. *PINN-LOCK* captures these characteristics, though the leaf-shaped velocity patterns appear slightly elongated, indicating minor structural discrepancies. Overall, spatial errors are primarily concentrated near the density interface, demonstrating the advantage of PINNs in physics-constrained interpolation (Fig. 3).

After 500 training epochs, the R^2 scores are 0.983 for horizontal velocity u, 0.925 for vertical velocity w, 0.724 for pressure p, and 0.992 for salinity s, indicating excellent agreement for most physical variables. The corresponding RMSE values are 0.130, 0.274, 0.525, and 0.089, respectively, with salinity achieving the highest predictive accuracy due to its relatively smooth spatial gradients. Pressure exhibits the lowest R^2, primarily due to its sensitivity to boundary conditions and the absence of certain pressure-related terms in the governing equations.

These results are obtained using only 1.3% of the total dataset, highlighting the model's robustness and generalization capability. Sensitivity analyses varying the number of hidden layers (4–8) and sampling ratios (0.5–2%) indicate that the physics-constrained loss is the primary factor governing prediction accuracy. Compared with purely data-driven interpolation methods, the PINN approach substantially reduces

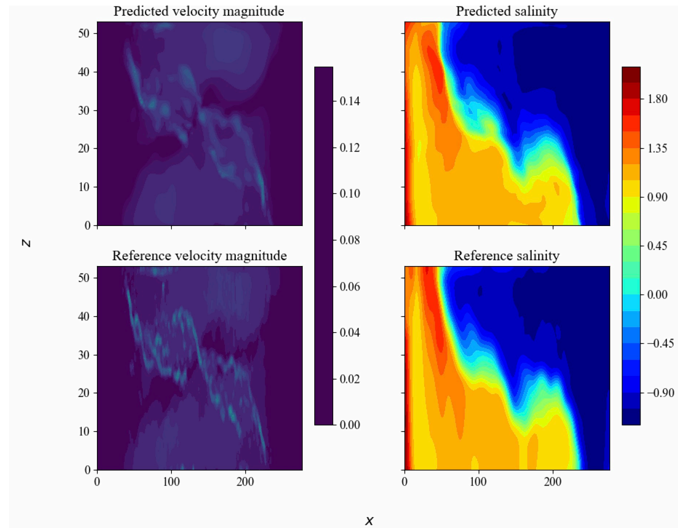

Fig. 4. Spatial distributions at $t = 300$.

prediction errors under sparse-data conditions, confirming its effectiveness in modeling buoyancy-driven flow dynamics.

B. PIV/PLIF Data

To validate *PINN-LOCK* on real experimental measurements, the model was trained using PIV/PLIF laboratory data. Unlike numerical simulations, experimental measurements contain inherent noise from optical uncertainties and particle tracking artifacts. PIV and PLIF provide only velocity and density fields; pressure serves as an inferred quantity through momentum equation constraints.

During mid-stage evolution ($t = 300$), complex vortex structures develop through Kelvin-Helmholtz instabilities. *PINN-LOCK* reproduces the spatial distribution of these features, though with smoother gradients reflecting viscous dissipation enforcement. The salinity field shows significant interfacial mixing accurately captured by the model (Fig. 4).

Quantitative assessment yields RMSE of 0.161 ($R^2 = 0.974$) for velocity, reflecting *PINN-LOCK*'s challenges with velocity reconstruction. Salinity achieves a small RMSE of 0.076 and $R^2 = 0.994$. Pressure cannot be validated against experimental data as PIV/PLIF provide no direct measurements; pressure is instead inferred through momentum equation residuals and only used during the training process.

Compared to SUNTANS results, experimental validation shows moderately higher errors: velocity RMSE increases from 0.130 to 0.161, while salinity improves slightly from 0.089 to 0.076. This degradation, particularly for velocity, reflects the quality difference between clean numerical data and noisy PIV measurements. Nevertheless, training on only 1.3% of available data points, *PINN-LOCK* successfully validates its practical applicability for real-world fluid mechanics applications, demonstrating superior generalization compared to purely data-driven methods.

IV. Discussion

We used only 1.3% of the data to reconstruct the full field through *PINN-LOCK*. To test this, we applied cubic interpolation using the same 1.3% of data. The results show that interpolation fails to handle such sparse data effectively.

Interpolation relies solely on existing data points and assumes smooth variation between them, which makes it unreliable for sparse datasets, especially in highly nonlinear or unsteady flows. It also cannot guarantee physical consistency, often producing results that violate fundamental laws such as mass or momentum conservation. In contrast, PINNs combine sparse observations with governing physical equations, allowing them to infer the flow field in unobserved regions while maintaining physical consistency. This makes PINNs particularly suitable for complex, non-linear, and non-stationary flow scenarios, providing more accurate and physically realistic predictions than interpolation.

V. Conclusions

This study examined density currents, a canonical example of self-driven flow, through an integrated approach combining physical experiments, theoretical analysis, numerical simulation, and deep learning. We developed *PINN-LOCK*, a Physics-Informed Neural Network framework that embeds physical laws into the learning process, providing an efficient and reliable method for predicting the evolution of unsteady, density-driven flows.

Using the SUNTANS numerical model, we achieved high-fidelity simulations of the Lock-exchange process, successfully reproducing counter-propagating flows, interfacial instabilities, vortex generation, and turbulent mixing. Sensitivity analyses confirmed that the front velocity scales with the square root of the initial density difference, in excellent agreement with theoretical predictions, thereby validating both the theoretical model and its numerical realization.

Building on these results, we implemented the Physics-Informed Neural Network approach as a surrogate modeling tool to enhance computational efficiency and generalization capability. By embedding high-resolution SUNTANS data and simplified governing equations into the network's loss function, the *PINN-LOCK* framework achieved rapid and physically consistent predictions of lock-exchange flow dynamics.

VI. Limitations and Future Work

Model extension: The present configuration assumes symmetric initial conditions with equal fluid volumes. Extending the model to asymmetric initial volumes would enable systematic exploration of how volume ratio influences front velocity, mixing behavior, and vortex evolution. **Three-dimensional modeling:** Extending the current two-dimensional (x, z) framework to fully three-dimensional (x, y, z) simulations will allow the investigation of spanwise instabilities, vortex stretching, and breakdown processes that cannot be captured in 2D. **Density contrast constraint:** Introducing the density difference $\Delta\rho$ explicitly into the loss function as a physical constraint could facilitate predictive modeling of flows under varying density contrasts.

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